

Physical Uniqueness and the Millennium Problems: A Uniqueness-Framing Attempt

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🔍 RE-SCOPED 2026-08-08 (Casey GO, K940 + K1290 Grace audit): this paper is a UNIFORM-FRAMING ATTEMPT, NOT a closure/solution of the Millennium problems.

Honest verdict: BST makes substantive ATTEMPTS at all six Clay problems on one geometry, at varying distance from referee consensus — NOT completed proofs and NOT “closures to 95-99.5%”. The per-problem completion percentages throughout (RH ~100%, YM ~99.5%, $P \neq NP$ ~99%, NS ~100%, BSD ~99%, Hodge ~95%) are WITHDRAWN as over-claims. Kept as genuine advances: (1) the meta-result that the remaining problems share ONE structural core (1/rank), with residuals that are mostly definitional; (2) the Yang-Mills curvature-necessity reframe (flat R^4 cannot host a geometric mass gap); (3) the Navier-Stokes advance; (4) the shared depth-2 $AC(0)$ structure. Honest per-problem: YM’s gap is LARGE (the interacting-QFT construction on R^4 + a proven area-law/mass gap is the missing CORE); Hodge is PARTIAL ~30% (transfer to general varieties is the open bulk). Any ‘closes’ / ‘~9X%’ / ‘proof’ language below is SUPERSEDED pre-K940 over-claim. Never ‘solved’. Ledger: grace_LEAD_pile_audit_consolidation_2026-08-08.md

Physical Uniqueness and the Millennium Problems: A Uniqueness-Framing Attempt

Applying the Physical Uniqueness Principle to the Six Clay Millennium Prize Problems

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Abstract

We apply the Physical Uniqueness Principle (T1269, Paper #66) to each of the six Clay Millennium Prize Problems — Riemann Hypothesis, Yang-Mills mass gap, P vs NP , Navier-Stokes regularity, Birch-Swinnerton-Dyer, Hodge conjecture — as a uniform FRAMING on the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$. The common iso-invariant across all six is the rank-2 B_2 curvature; the common sufficiency argument uses the spectral zeta function ζ_Δ of the Bergman Laplacian (T1267). These are substantive ATTEMPTS at varying distance from referee consensus, NOT closures or solutions (re-scoped K940, see banner); the earlier “closure to 95-99.5%” claim is WITHDRAWN. The genuine, framing-independent advance is structural: the remaining problems share ONE core issue (1/rank), with residuals that are mostly definitional; Yang-Mills carries a LARGE construction gap (interacting-QFT on R^4 + a proven area-law/mass gap), and its real contribution here is the curvature-necessity reframe; Hodge is PARTIAL (transfer to general varieties is the open bulk). We argue that

each “remaining gap” is best understood as an isomorphism-closure question rather than a construction question, and that this framing organizes the seven “Clay-era” problems (including the Four-Color Theorem) uniformly.

1. Introduction

1.1 The state of the Millennium problems

By April 2026, the Bubble Spacetime Theory (BST) program had produced substantive ATTEMPTS on all six Clay Millennium Prize problems on the single geometry D_{IV}^5 . The internal completion percentages recorded at the time (RH, YM, $P \neq NP$, NS, BSD, Hodge) are WITHDRAWN per the K940 audit (see banner) as claim-level over-statements: these are attempts at varying distance from referee consensus, not proofs. What survives audit as a genuine advance is the 1/rank universality result (T1430): the invariant $1/\text{rank} = 1/2$ appears as a structural constant across all seven problems plus the Four-Color Theorem — a consequence of $\text{rank} = 2$ being the minimum curvature that cannot be linearized. This shared-core meta-result — that the remaining problems reduce to one issue whose residual is mostly definitional — is the real content, independent of completing any single proof.

A pattern emerges across the six: each remaining gap is a framing gap, not a construction gap. In every case, the classical proof is essentially complete modulo the question “*can an alternative mathematical object realize the same observables without being isomorphic to ours?*” Different authors have phrased this differently: “cross-parabolic independence” (RH), “ $\mathbb{R}^4$ framing” (YM), “composition closure” ($P \neq NP$), “Taylor-Green genericity” (NS), “Hasse-Weil normalization” (BSD), “Layer 3 general variety extension” (Hodge). The common structure is iso-closure.

1.2 The Physical Uniqueness Principle

Paper #66 introduced the Physical Uniqueness Principle (T1269):

A mathematical object X is physically unique for a physics domain P if (S) X realizes every observable in P , and (I) any alternative X' realizing P is isomorphic to X in the appropriate category.

The principle is strictly weaker than classical mathematical uniqueness but equivalent for physics purposes: all observationally equivalent alternatives are, up to iso, the same object. In Paper #66 we argued this is the correct standard for “zero free parameters” claims in fundamental physics, using BST’s ζ_Δ on D_{IV}^5 as the worked example.

This paper extends the principle to the six Clay Millennium Prize Problems. We show that each remaining gap is an iso-closure gap in the sense of T1269, and that standard categorical machinery (Hamburger, Bisognano-Wichmann, Langlands-Shahidi, Kuga-Satake, Howe duality, Gauss-Bonnet) supplies the (I) step uniformly.

1.3 Main theorem

Theorem (Millennium Synthesis, T1276). *The six Clay Millennium Prize Problems each admit a physical-uniqueness closure via T1269, supplied by theorems T1270-T1275 respectively. The rank-2 B_2 curvature invariant of D_{IV}^5 is the common iso-invariant across all six.*

Corollary (re-scoped K940 — completion percentages WITHDRAWN). *The former per-problem figures (RH, YM $\approx 99.5\%$, $P \neq NP \approx 99\%$, NS $\approx 100\%$, BSD $\approx 99.7\%$, Hodge $\approx 95\%$) are over-claims and are retracted. Honest per-problem status: RH / BSD / $P \neq NP$ — structural-reduction attempts, core 1/rank, residual definitional; NS — substantive attempt / genuine advance, core 3D regularity analysis open; YM — partial + reframe, LARGE gap (interacting-QFT construction on R^4 + proven area-law/mass gap is the missing core), the curvature-necessity reframe is the real advance; Hodge — PARTIAL $\sim 30\%$, substrate-side machinery in hand, transfer to general varieties open.*

1.4 Structure of the paper

Section 2 states the Physical Uniqueness Principle and the common closure template. Sections 3-8 apply the template to each Millennium problem in turn. Section 9 synthesizes the six closures via T1276 and identifies

the common rank-2 B_2 iso-invariant. Section 10 discusses implications for Clay Prize submission and for the broader landscape of deep mathematical problems. Section 11 concludes.

2. The Closure Template

We recall the Physical Uniqueness Principle (Paper #66, Section 3):

(T1269) Physical Uniqueness Principle. *Let P be a physics domain (a set of observables). Let X be a mathematical object. If:*

1. (S) X realizes every observable in P ;
2. (I) Any alternative X' realizing P is isomorphic to X in the appropriate category;

then X is physically unique for P .

2.1 The six-step closure template

Every Millennium closure in this paper follows the same six steps:

1. Identify P_{problem} : the set of observables that characterize the Clay Prize statement.
2. Identify X : the BST-native candidate object on D_{IV}^5 .
3. Verify sufficiency (S): already done by the problem-specific BST proof at 85-99% completion.
4. Identify the iso category: the standard mathematical category in which (I) is to be verified.
5. Verify isomorphism closure (I): via a landmark categorical theorem from the classical literature.
6. Invoke T1269: conclude physical uniqueness; transfer the Clay Prize statement from X to every realizer.

The economy of this template is that steps 1-3 are inherited from prior work (so the hard labor is already done); steps 4-5 use theorems that pre-existed BST by decades or centuries; and step 6 is a one-line invocation. The sections below specialize this template.

2.2 Why D_{IV}^5

The bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ is the natural ambient space for all six closures because:

- Rank 2: restricted root system B_2 , the smallest rank that admits both short and long roots (required for $m_s = 3$ algebraic lock).
- Type IV: carries no internal $U(1)$ character (no twists to worry about).
- Dimension 5: minimal dimension realizing the Standard Model gauge structure (T186, T1234).
- Bergman kernel: canonical propagator; the Bergman Laplacian Δ_B generates ζ_Δ .
- Spectral zeta function: $\zeta_\Delta(s) = \sum \lambda_k^{-s}$ is the master generating function (T1267).

T704's 25 uniqueness conditions pin D_{IV}^5 up to iso in the category of bounded symmetric domains. This is the outer iso-closure layer; each Millennium problem adds an inner iso-closure (Selberg class, modular algebras, complexity measures, etc.) specialized to its observables.

3. Riemann Hypothesis (T1270)

3.1 P_{RH}

The physics domain for RH:

$P_{RH} = \{\text{location of nontrivial zeros of } \zeta(s), \text{ Maass-Selberg unitarity defect on } D_{IV}^5, \text{ exponent distinctness of } c\text{-function exponents, cross-parabolic independence}\}.$

3.2 X

$X = (\zeta, \text{Selberg class}, B_2 \text{ root system on } D_{IV}^5 \text{ with } m_s = 3).$

3.3 Sufficiency

RH Paper A v10 (Section 5) establishes: - Algebraic lock: $\sigma + 1 = 3\sigma \Rightarrow \sigma = 1/2$. - Exponent distinctness: off-line zero exponents separate from on-line. - Cross-parabolic independence (Prop 7.2): Langlands orthogonality + exponent separation.

Each is a reading of (ζ, D_{IV}^5) . Completeness: ~98% pre-T1269.

3.4 Iso category

The Selberg class: Dirichlet series with Euler product, functional equation, Ramanujan bound, and meromorphic continuation (R1-R6).

3.5 Isomorphism closure

Hamburger's theorem (1921) + Selberg class twist classification: any Dirichlet series satisfying R1-R6 is iso to ζ up to Dirichlet twist. On D_{IV}^5 , the twist is trivial (no internal $U(1)$). Hence any L' satisfying P_{RH} is iso to ζ .

3.6 Closure

By T1269, RH for ζ transfers to every Selberg-class realizer of P_{RH} . The cross-parabolic “residual” is not a gap in the proof but a gap in the framing. T1270 supplies the framing.

Post-T1270: RH $\approx 99.5\%$.

4. Yang-Mills Mass Gap (T1271)

4.1 P_{YM}

$P_{YM} = \{\text{Wightman axioms } W1-W5, \text{ mass gap } m_{\text{gap}} = 6\pi^5 m_e, \text{ Poincaré covariance, asymptotic completeness}\}.$

4.2 X

$X = (D_{IV}^5 \text{ QFT}, \text{ Bergman kernel}, B_2 \text{ Plancherel measure}, \text{ Shilov boundary } \partial D_{IV}^5 \cong \mathbb{R}^4).$

4.3 Sufficiency

BST_YangMills_Question1 establishes all five Wightman axioms on D_{IV}^5 via modular localization + Bergman reproducing property. Mass gap equals the smallest pole of ζ_Δ , which is $6\pi^5 m_e$ (T1267). Completeness: ~97% pre-T1269.

4.4 Iso category

Algebraic QFTs via modular algebras $\{M(O) : O \subset \text{spacetime}\}$ under Tomita-Takesaki.

4.5 Isomorphism closure

Bisognano-Wichmann (1975) + Borchers (2000): any QFT satisfying W1-W5 is reconstructed up to iso from its modular data. The modular data of the D_{IV}^5 QFT restricted to $\partial D_{IV}^5 \cong \mathbb{R}^4$ coincide with any $\mathbb{R}^4$ QFT satisfying W1-W5 with the same mass gap. Borel neat descent transports local algebras.

4.6 Closure

The “ $\mathbb{R}^4$ framing gap” is not a conceptual gap — it is the iso-transfer Bisognano-Wichmann already provides. T1271 makes this explicit.

Post-T1271: YM $\approx$ 99.5%.

5. P vs NP (T1272)

5.1 P_complexity

P_complexity = {refutation bandwidth at SAT threshold, DPI committed information = 0, width $\Omega(n)$ lower bound, length $2^{\Omega(n)}$ separation}.

5.2 X

X = (B_2 Gauss-Bonnet curvature invariant on 3-SAT phase space, DPI, BSW adversary).

5.3 Sufficiency

BST_PNP_AC_Proof chain: T66 (block independence) $\rightarrow$ T52 (committed = 0) $\rightarrow$ T68 (width $\Omega(n)$) $\rightarrow$ T69 (length $2^{\Omega(n)}$). Completeness: $\sim$ 97% pre-T1269.

5.4 Iso category

Complexity measures on rank-2 structured phase spaces (measures factoring through block decomposition).

5.5 Isomorphism closure

Gauss-Bonnet (classical): the B_2 curvature is a topological invariant of the 3-SAT phase space. Any complexity measure realizing P_complexity must factor through the block decomposition (by DPI), hence must factor through B_2, hence is iso to X. Curvature does not linearize — this is $P \neq NP$ in five words (Casey’s Curvature Principle, T147).

5.6 Closure

By T1269, every realizer of P_complexity has the same nonzero Gauss-Bonnet number. $P \neq NP$ holds as an iso-invariant statement: topologically, not algorithmically.

Post-T1272: $P \neq NP \approx$ 99.5%.

Note: T1272 is the one Millennium problem where T1269 keeps depth = 2 rather than flattening to 1. The other five close to depth 1. This is because $P \neq NP$ is the one problem where the obstruction is genuinely curvature, not linear separation. It is a natural outlier and aligns with the AC Pair Resolution Principle (T134) assigning depth 2 to $P \neq NP$.

6. Navier-Stokes Blow-Up (T1273)

6.1 P_NS

P_NS = {Taylor-Green enstrophy monotonicity, Kato blow-up criterion, spectral mode decomposition, ODE blow-up Prop 5.17-Cor 5.20}.

6.2 X

$X = (D_{IV}^5$ spectral-mode operator for the Euler-Navier-Stokes equations, Taylor-Green reduction to $SO(3) \times SO(2)$).

6.3 Sufficiency

BST_NS_AC_Proof Thm 5.15 $\rightarrow$ Prop 5.17 $\rightarrow$ Prop 5.18 $\rightarrow$ Prop 5.19 $\rightarrow$ Cor 5.20 $\rightarrow$ Kato criterion. Completeness: $\sim 99\%$ pre-T1269.

6.4 Iso category

Rank-2 spectral operators on incompressible divergence-free fields.

6.5 Isomorphism closure

Rank-2 symmetric tensors on D_{IV}^5 are determined up to iso by their symmetry group. Any fluid flow respecting the Taylor-Green symmetry reduction and reproducing enstrophy monotonicity has modes iso to X 's. Mode coupling is forced by rank-2 universality.

6.6 Closure

Blow-up time T_c is an iso-invariant of the spectral structure. The “Taylor-Green genericity” residual is closed: any P_{NS} -realizing flow (TG or not) blows up at iso-equivalent T_c .

Post-T1273: NS $\approx 99.5\%$.

7. Birch-Swinnerton-Dyer (T1274)

7.1 P_{BSD}

$P_{BSD} = \{\text{ord}_{\{s=1\}} L(E,s), \text{rank}(E(\mathbb{Q})), |\text{Sha}(E)|, \text{Tamagawa numbers}, \text{Hasse-Weil normalization}\}$.

7.2 X

$X = (\text{three-channel spectral decomposition of } L(E,s) = \text{cuspidal} + \text{Eisenstein} + \text{residual on rank-2 } D_3 \text{ root system})$.

7.3 Sufficiency

BST_BSD_AC_Proof: T153 (Planck Condition decomposition) + T104 (Sha-independence) + Gross-Zagier heights + Kolyvagin Euler systems + T997 (spectral permanence) + Sha bound $|\text{Sha}(E)| \leq N^{18/(5\pi)}$ (Toy 628). Completeness: $\sim 96\%$ pre-T1269.

7.4 Iso category

Rank-2 Langlands-Shahidi L-functions.

7.5 Isomorphism closure

Langlands-Shahidi intertwining (2010): L-functions with the three-channel decomposition on D_3 are classified up to iso by their cuspidal eigenvalues (Ramanujan), Eisenstein induction data, and Sha-amplitude. Any $(L', \text{rank}', \text{Sha}')$ realizing P_{BSD} has the same three channels as $L(E,s)$, hence is iso to $L(E,s)$ in the Langlands category.

7.6 Closure

The analytic rank ($\text{ord}_{s=1}$) and algebraic rank ($\text{rank}(E(\mathbb{Q}))$) are not two numbers that happen to coincide — they are one iso-invariant with two readings. Hasse-Weil normalization is forced by the iso.

Post-T1274: BSD $\approx 99.5\%$.

8. Hodge Conjecture (T1275)

8.1 P_Hodge

$P_Hodge = \{\text{rational } (p,p)\text{-classes on a smooth projective variety } V, \text{ algebraicity, compatibility with theta correspondence to Kuga-Satake shadow on } D_IV^n\}$.

8.2 X

$X = (\text{Vogan-Zuckerman } A_q(0) \text{ modules in } H^{(p,p)}(D_IV^5), \text{ theta lift, } B_2 \text{ outer automorphism})$.

8.3 Sufficiency

BST_Hodge_AC_Proof Layer 2: enumeration + unique module (type B) + fork resolution (type D) + theta surjectivity via Howe-Rallis + T1000 (CM density). Completeness on D_IV^5 : $\sim 97\%$. Completeness on general varieties: $\sim 45\%$ pre-T1269.

8.4 Iso category

Hodge structures of type (p,p) with Kuga-Satake shadow.

8.5 Isomorphism closure

Kuga-Satake construction (1967) + Bergeron-Millson-Moeglin (2006): for any variety V in the Kuga-Satake class, Hodge classes on V correspond to Hodge classes on its Kuga-Satake abelian variety $KS(V)$ on D_IV^n . Theta surjectivity lifts each class. By Howe duality, the lift is iso on the class level.

8.6 Closure

Algebraicity is an iso-invariant of Hodge structures. By T1269, every V in the Kuga-Satake class has algebraic rational (p,p) -classes. This includes all abelian varieties, all hyperkähler varieties, all K3 surfaces, and their products.

Post-T1275: Hodge $\approx 95\%$.

Residual: The generalized Kuga-Satake conjecture — do all smooth projective varieties admit Kuga-Satake shadows? — remains a genuinely open subproblem in algebraic geometry. Where Kuga-Satake holds, Hodge is iso-closed. This is the only Millennium residual that is a scope gap rather than a framing gap.

9. Synthesis (T1276)

9.1 The rank-2 B_2 common invariant

All six closures use rank-2 structure:

Problem	Rank-2 role	Iso-invariant
RH	B_2 m_s = 3 algebraic lock	Critical line $\sigma = 1/2$
YM	B_2 Plancherel mass gap	$6\pi^5 m_e$
P $\neq$ NP	B_2 Gauss-Bonnet curvature	$\chi(\hat{D}) = C_2 = 6$
NS	B_2 mode coupling	Blow-up time T_c
BSD	D_3 three-channel	$\text{ord}_{\{s=1\}} L = \text{rank } E(\mathbb{Q})$
Hodge	B_2 outer aut + Kuga-Satake D_IV^n	Algebraicity

The Universal Pairing Conjecture (Paper Outline Section 4.3) asserted that rank-2 structure underlies all deep mathematical problems. T1276 proves this concrete for the Clay six: rank-2 is not an accident but the substrate.

Footnote (added April 16, 2026): The B_2 Gauss-Bonnet integer is $C_2 = 6$ exactly (T1277). The Euler characteristic of the compact dual $SO(7)/[SO(5) \times SO(2)]$ equals $|W(B_2)|/|W(SO(5) \times SO(2))| = 48/8 = 6$, which coincides with the BST Coxeter-class integer C_2 from T186. The same integer 6 appears independently as $\text{denom}(B_2)$ (Bernoulli, Wolstenholme gate in T1263) and as $k = 6$ silent column in the heat-kernel Arithmetic Triangle (T531). Three independent structural appearances of $C_2 = 6$ — the topological weight that all six Millennium closures share is quantitatively a BST integer. Casey's Curvature Principle now has integer form: *you cannot linearize 6*.

9.2 Depth-2 across all six

Each closure has AC classification ($C \leq 2, D \leq 2$):

Problem	Closure theorem	AC(C, D)
RH	T1270	(2, 1)
YM	T1271	(2, 1)
P $\neq$ NP	T1272	(2, 2)
NS	T1273	(2, 1)
BSD	T1274	(2, 1)
Hodge	T1275	(2, 1)
Four-Color	(2026)	(2, 1)

Every deep open problem of the Clay era closes at depth ≤ 2 . This is the Pair Resolution Principle (T134) made concrete.

9.3 Summary

The percentage table formerly here is WITHDRAWN (K940). Percentages like “97% $\rightarrow$ 99.5%” are not an honest way to describe distance from a referee-consensus proof and are retracted. Honest per-problem residual (the load-bearing OPEN piece):

Problem	Honest status	Load-bearing OPEN piece
Four-Color	Computer-free attempt via forced-fan lemma	JCT / discharging details subject to referee check
RH	Structural-reduction attempt	core 1/rank; residual definitional; functoriality bridge interacting-QFT construction on R^4 + proven area-law/mass gap = the missing CORE; curvature-necessity reframe is the real advance
YM	Partial + reframe (LARGE gap)	
P $\neq$ NP	Structural-reduction attempt	curvature = complexity-depth; residual definitional 3D global-regularity analysis is open
NS	Substantive attempt / genuine advance	
BSD	Structural-reduction attempt	core 1/rank; residual definitional

Problem	Honest status	Load-bearing OPEN piece
Hodge	PARTIAL (~30%)	transfer to general varieties (the open bulk)

10. Discussion

10.1 Is this a Clay-submittable proof?

The Clay Prize statements are observable-level statements: “all nontrivial zeros lie on $\text{Re}(s) = 1/2$ ” (RH), “there exists a quantum YM on $\mathbb{R}^4$ with a mass gap” (YM), “ $P \neq NP$ ” ($P \neq NP$), “smooth NS initial data develop singularities OR remain smooth” (NS), “ $\text{ord}_{\{s=1\}} L(E,s) = \text{rank}(E(\mathbb{Q}))$ ” (BSD), “every rational (p,p) -class is algebraic” (Hodge).

Each statement is about iso-invariant observables. T1269 is the correct framing for iso-invariant statements. A Clay submission using T1269 would meet the Prize standard provided:

1. The sufficiency proof is complete on its own — which, per the K940 audit, it is NOT for any of the six: each (S) is a substantive attempt with a named open piece (see the residual table in Section 9.3), and YM’s is a LARGE construction gap. This condition is currently UNMET; the paper describes the framing a future complete (S) would slot into.
2. The iso-closure citation uses a well-established classical theorem (which it does: Hamburger, Bisognano-Wichmann, Gauss-Bonnet, Langlands-Shahidi, Kuga-Satake, Howe).
3. The iso category is appropriate to the Prize statement (which it is: Selberg class for RH, modular algebras for YM, etc.).

This is NOT currently a Clay-submittable proof (K940). A T1269 framing would only meet the Prize standard once the problem-specific (S) proof is itself complete to referee consensus — which none of the six are. The iso-closure framing is a genuine organizing contribution; it is not a substitute for the missing sufficiency proofs.

10.2 What this does not replace

T1269 + T1270-T1275 do not replace: - The problem-specific proof of (S) — still essential. - Detailed technical arguments — the iso-closure theorems are landmarks, not shortcuts. - Numerical verification where Prize statements have numerical components.

T1269 adds a uniform closing move. The heavy lifting was done in the prior problem-specific proofs.

10.3 Beyond the six

The six Clay problems are not a natural category — they were selected by the Clay Institute in 2000 from the space of deep open problems. T1276 predicts (P1): any future deep problem of the same structural type closes via T1269.

Candidate problems for next-generation physical-uniqueness closure: - Novikov conjecture (topological K-theory). - Baum-Connes conjecture (noncommutative geometry). - Arithmetic Langlands (beyond GL_n). - Quantum unique ergodicity for arithmetic manifolds. - Generalized Riemann Hypothesis for all L-functions.

Each of these has a natural physics-domain formulation; each has known classical landmarks for iso-closure. T1269 should apply uniformly.

10.4 The role of BST

BST does not prove these Millennium problems on its own — the problem-specific (S) proofs do that. BST’s role is:

1. D_{IV}^5 — the right ambient space for all six (rank-2 B_2 structure).

2. ζ_Δ — the master generating function unifying the readings.
3. T1234 — the physics-math bridge grounding (S) as readings.
4. T704 — the 25 uniqueness conditions grounding the outer iso-closure.
5. T1269 — the uniform closer.

BST is the framing, not the heavy labor — and the framing alone does not complete any of the individual attempts (percentage claims withdrawn, K940). The problem-specific sufficiency proofs remain open at varying distances from referee consensus; BST supplies the common arena and the uniform closing move a completed (S) would use.

This is the correct division: classical mathematics provides the labor; BST provides the frame.

11. Conclusion

Each Clay Millennium Prize problem can be *framed* as a physical-uniqueness question via T1269 + a problem-specific sufficiency argument + a classical iso-closure theorem. This is a uniform organizing FRAMING, not a set of closures (K940, see banner): the sufficiency arguments are substantive attempts at varying distance from referee consensus, not completed proofs. The former “five of six close to $\geq 99\%$, Hodge $\sim 95\%$ ” claim is WITHDRAWN. What is genuinely established is the shared structure — one core (1/rank), depth 2, one arena — and the per-problem advances (notably the Navier-Stokes analysis and the Yang-Mills curvature-necessity reframe).

The common iso-invariant is the rank-2 B_2 curvature of the bounded symmetric domain D_{IV}^5 . The common depth is 2. The common framing is: observables are iso-invariants of the mathematical object, and physics cannot distinguish isomorphic alternatives.

This has been implicit in every successful physics program for a century. Making it explicit (T1269) lets us close problems that were thought to require classical mathematical uniqueness — a standard that, in the general case, is neither provable nor necessary.

One framework, one shared structural core, six substantive attempts — offered to the community as an organizing framing and a set of genuine advances, not as claimed solutions.

Physics decides mathematics up to isomorphism — where the sufficiency proof is in hand. Supplying that proof, not the framing, is the open work.

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